

$$g(x) = \frac{x^2 - 4x + 6}{x^2 - 4}$$

- a. Domain = $\{x : x \neq -2, x \neq 2\}$
- b. Intercepts $(0, -3/2)$, no x-intercepts
- c. Asymptotes

1. Horizontal: $y = 1$ since $\lim_{x \rightarrow \infty} \frac{x^2 - 4x + 6}{x^2 - 4} = 1$

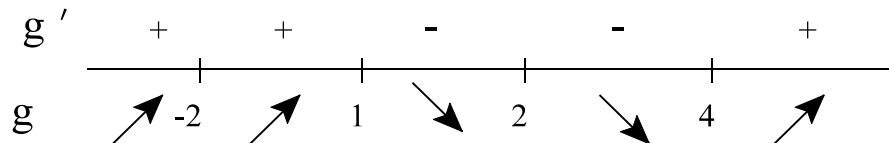
2. Vertical:

$$x = 2 \text{ since } \lim_{x \rightarrow 2^+} \frac{x^2 - 4x + 6}{x^2 - 4} = +\infty, \lim_{x \rightarrow 2^-} \frac{x^2 - 4x + 6}{x^2 - 4} = -\infty$$

$$x = -2 \text{ since } \lim_{x \rightarrow -2^+} \frac{x^2 - 4x + 6}{x^2 - 4} = -\infty, \lim_{x \rightarrow -2^-} \frac{x^2 - 4x + 6}{x^2 - 4} = +\infty$$

d. $g'(x) = \frac{4(x^2 - 5x + 4)}{(x^2 - 4)^2} \Rightarrow$ C.N. $x = -2, x = 1, x = 2, x = 4$

e.



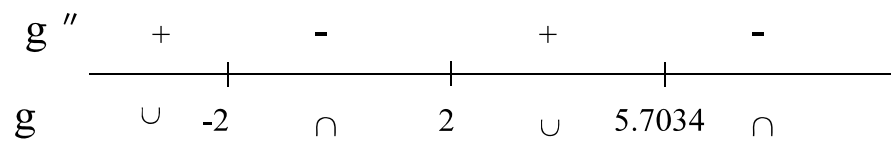
Increasing on $(-\infty, -2) \cup (-2, 1) \cup (4, \infty)$

Decreasing on $(1, 2) \cup (2, 4)$

- f. $P(1, -1)$ is a relative maximum point on the graph
 $P(4, 1/2)$ is a relative minimum point of the graph

g. $g''(x) = -\frac{4(2x^3 - 15x^2 + 24x - 20)}{(x^2 - 4)^3} \Rightarrow$ 2nd C.N. $x = -2, x = 2, x = 5.7034$

h.



Concave up on $(-\infty, 2) \cup (2, 5.7034)$

Concave down on $(-2, 2) \cup (5.7034, \infty)$

i. Inflection points: $(5.7034, 0.5509)$

j. Values

x	-7	-6	-5	-4	-3	-1	0	1	3	4	5	6	7
y	1.84	2.06	2.43	3.17	5.4	-3.66	-1.5	-1	0.6	0.5	0.52	0.56	0.6

k.

